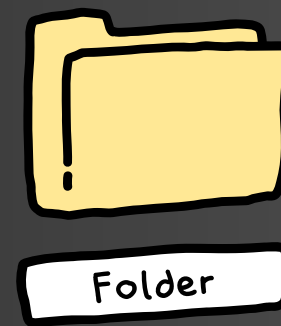




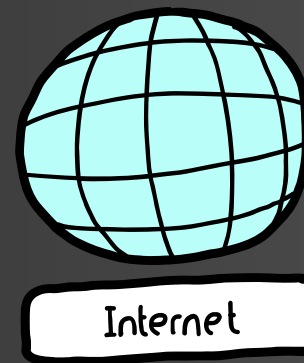
DNS Setup



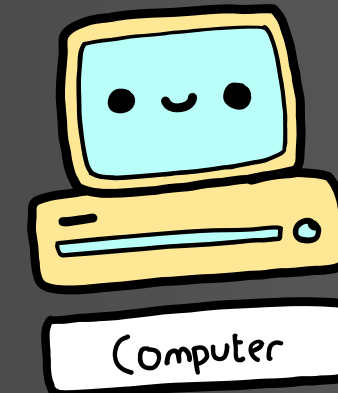
Folder



e-mail



Internet



Computer

Topics

- Apache webserver Setup
- DNS Setup
- Custom Domain Name setup

What is DNS?





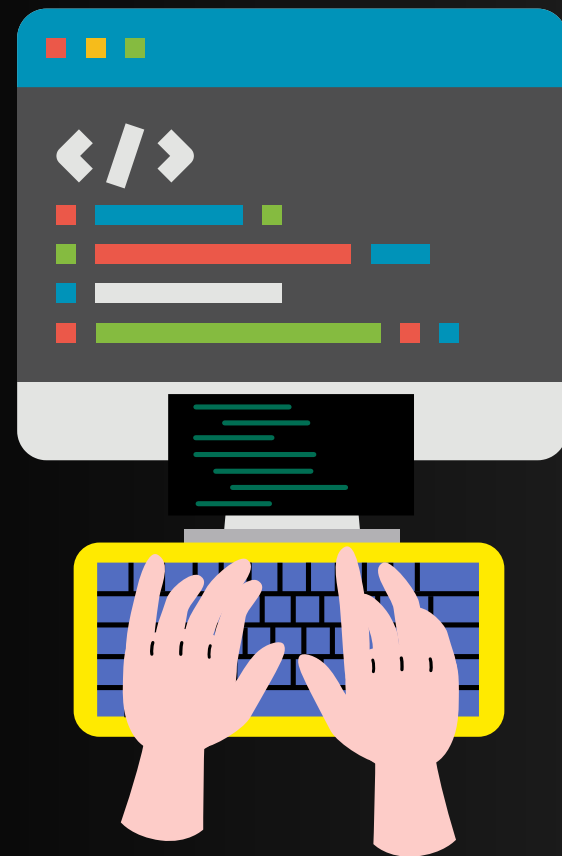
DNS, or Domain Name System, is the internet service that translates human-friendly domain names like **www.example.com into machine-readable IP addresses.**

How DNS works?



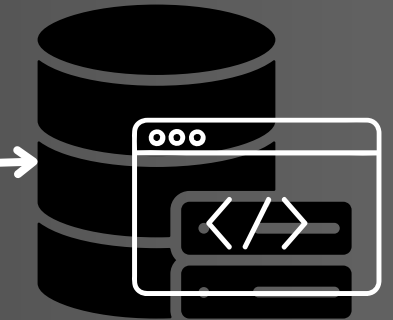


www.google.com



```
www.google.com 192.168.0.1
www.example.com 192.168.0.2
www.helloo.com 192.168.0.3
```

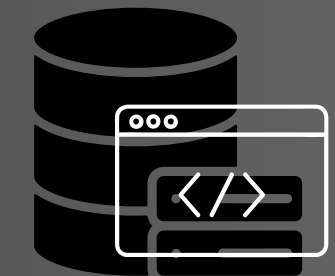
DNS Server



192.168.0.1



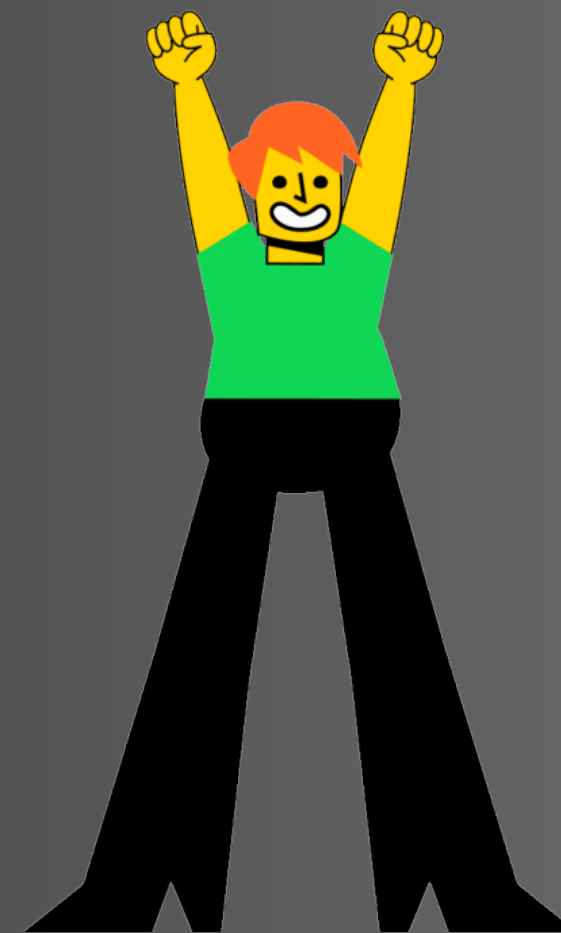
192.168.0.2



192.168.0.3



Let's setup our own WebServer



Installation (CentOS or RedHAT)

Package we need to install

- `sudo yum install httpd`
- `systemctl start/stop/status httpd` (httpd is the process or service name)

Enable the Service in Firewall

`firewall-cmd --add-service=http --permanent`

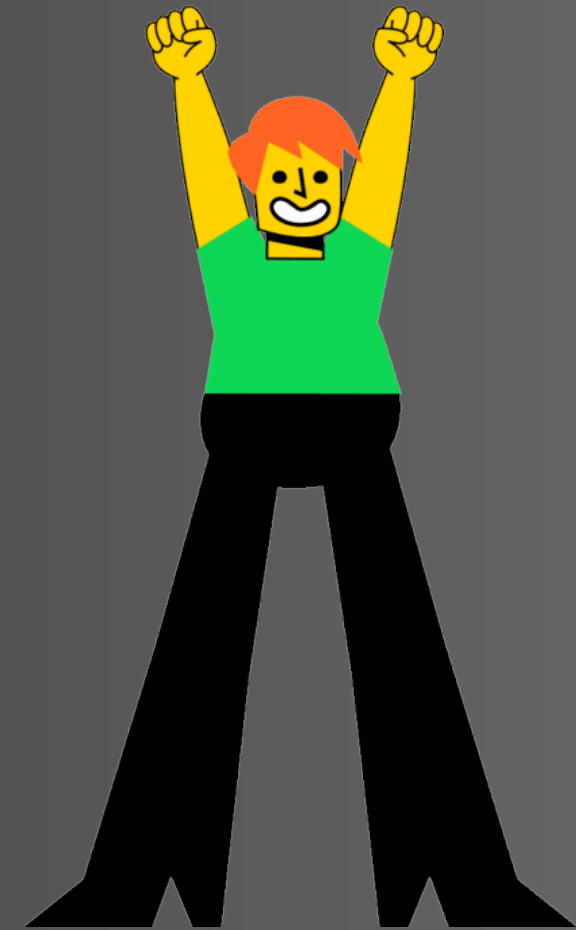
`firewall-cmd --reload`

Webserver config file under

- `/var/www/html/index.html`
- `/etc/httpd/conf/httpd.conf`



Let's setup our own DNS Server



Installation (CentOS or RedHAT)

Package we need to install for the DNS is BIND (Berkeley Internet Name Domain)

- `sudo yum install bind bind-utils`
- `systemctl start/stop/status named` (named is the process or service name)

Enable the Service in Firewall

`firewall-cmd --add-service=dns --permanent`

`firewall-cmd --reload`

DNS config file under

- `/etc/named.conf`

Directory where all the zone files are present where you define hostname to IP etc

- `/var/named`

DNS Translate

- **Hostname to 192.168.1.2 (Called A Record)**
- **192.168.1.2 to hostname (Called PTR Record)**
- **Hostname to hostname (Called CNAME Record)**

Zones Files

- **Forward zone - resolve Domain to IP**
- **Reverse zone - resolve IP to Domain**

Here's how to configure your DNS settings on different operating systems:

- **Windows:** Go to Control Panel > Network and Internet > Network and Sharing Center > Change adapter settings. Right-click your network connection, select Properties, then select Internet Protocol Version 4 (TCP/IPv4) or Version 6 (TCP/IPv6) and click Properties. Here, you can set your preferred DNS server.
- **macOS:** Go to System Preferences > Network, select your network interface, click Advanced, and go to the DNS tab. You can add your DNS server here.
- **Linux:** This depends on your distribution and network manager, but typically you can edit `/etc/resolv.conf` directly or configure through network management tools (like NetworkManager) to add your DNS server.